The Verification Tax: Fundamental Limits of AI Auditing in the Rare-Error Regime

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Abstract

The most cited calibration result in deep learning—post-temperature-scaling ECE of 0.012 on CIFAR-100 [Guo et al., 2017]—is below the statistical noise floor. We prove this is not a failure of the experiment but a law: the minimax rate for estimating calibration error with model error rate ε is $\Theta((L\varepsilon/m)^{1/3})$, and no estimator can beat it. This "verification tax" implies that as AI models improve, verifying their calibration becomes fundamentally harder—with the same exponent in opposite directions. We establish four results that contradict standard evaluation practice: (1) self-evaluation without labels provides *exactly zero* information about calibration, bounded by a constant independent of compute; (2) a sharp phase transition at $m \cdot \varepsilon \approx 1$ below which miscalibration is undetectable; (3) active querying eliminates the Lipschitz constant, collapsing estimation to detection; (4) verification cost grows exponentially with pipeline depth at rate L^K. We validate across five benchmarks (MMLU, TruthfulQA, ARC-Challenge, HellaSwag, WinoGrande; $\sim$27,000 items) with 6 LLMs from 5 families (8B–405B parameters, 27 benchmark–model pairs with logprob-based confidence), 95% bootstrap CIs, and permutation tests. Self-evaluation non-significance holds in 80% of pairs. Across frontier models, 23% of pairwise comparisons are indistinguishable from noise, implying that credible calibration claims must report verification floors and prioritize active querying once gains approach benchmark resolution.

1 Introduction

The base rate problem—that rare conditions are hard to detect—is among the oldest results in statistics. We show that AI auditing faces the same fundamental barrier. The "disease" is miscalibration: the gap between a model's stated confidence and its actual accuracy. As errors become rare, miscalibration becomes a rare condition, and the verification cost grows without bound. We call this the *verification tax*. In one line: better models are harder to audit.

This result completes a programme initiated by Sun et al. [2023], who established the $O(n^{-2/3})$ upper bound for ECE estimation (NeurIPS 2023 Spotlight), and extended by Futami and Fujisawa [2024], who generalized the rate information-theoretically (NeurIPS 2024). Both left two questions open: (1) Is $n^{-1/3}$ the optimal rate? (2) How does the model's error rate ε affect verification difficulty? We resolve both. The lower bound (Theorem 2) matches the upper bound (Theorem 3), establishing $(L\varepsilon/m)^{1/3}$ as the minimax rate. The ε-dependence—absent from all prior work—reveals a phase transition, a scaling duality, and a verification horizon.

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The tax is a law: no estimator can beat $\Theta((L\varepsilon/m)^{1/3})$. Under meaningful verification ($\delta = \Theta(\varepsilon)$), sample complexity grows as $\Omega(1/\varepsilon^2)$. Capability and verifiability are governed by the same exponent in opposite directions (Theorem 4).

But the rate is the scaffolding. The paper's contributions are four results that contradict standard assumptions in AI evaluation [Liang et al., 2023, Zheng et al., 2023, Chiang et al., 2024]:

Surprise 1: Self-evaluation is bounded by a constant (§3). Any estimator that does not use ground-truth labels—including LLM-as-Judge, self-consistency, chain-of-thought verification, and all model-derived signals—has worst-case calibration error $\geq 1/2$ for $L \geq 1$. The bound is independent of the number of queries. White-box model access does not help unless it identifies a low-dimensional parametric family.

Surprise 2: There is a sharp phase transition (§4). Below $m \cdot \varepsilon \approx 1$, miscalibration is undetectable by *any* method. We show that most published safety benchmark improvements fall below the verification floor (Table 4), and 65% of MMLU per-subject rankings are noise.

Surprise 3: Active querying eliminates L (§5). When the auditor chooses which inputs to test, the minimax rate improves to $\Theta(\sqrt{\varepsilon/m})$—the Lipschitz constant disappears entirely. We validate this on production LLMs: all three models collapse to the same active rate despite $\hat{L} \in [1.4, 2.5]$.

Surprise 4: Composition is exponential (§6). A K-component AI pipeline has verification cost $\Omega(L^K \cdot \varepsilon/\delta^3)$. A 10-step agent loop with $L = 2$ costs $1{,}024\times$ a single-step model. This applies to chain-of-thought, retrieval-augmented generation, and agentic systems.

Together, these results imply that the dominant practices in AI evaluation—passive benchmarking, self-evaluation, and end-to-end testing of complex pipelines—operate in provably information-less regimes. We demonstrate this empirically and show that every frontier model exceeds the verification horizon (§7).

2 The Verification Tax

Setup. Let f be a classifier with confidence $p(x) \in [0, 1]$. The calibration gap is $\Delta(p) = \eta(p) - p$ where $\eta(p) = \mathbb{E}[Y{=}\hat{y} \mid p(X){=}p]$. The binned ECE is $\mathrm{ECE}_B = \sum_b (n_b/n)|\bar{\Delta}_b|$. We consider L-Lipschitz calibration functions $\mathcal{F}(L) = \{\Delta : |\Delta(p_1) - \Delta(p_2)| \leq L|p_1 - p_2|\}$ and the minimax risk $R^*(m, \varepsilon, L) = \inf_{\widehat{\mathrm{ECE}}} \sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}[|\widehat{\mathrm{ECE}} - \mathrm{ECE}|]$, where $\varepsilon = \Pr[Y \neq \hat{y}]$ is the error rate.

Theorem 1 (Le Cam Lower Bound)**.** *$R^*(m, \varepsilon, L) \geq c_1\sqrt{\varepsilon/m}$, where $c_1 \geq 1/(4\sqrt{2})$.*

Proof sketch. Construct P_0 (calibrated, $\mathrm{ECE} = 0$) and P_1 (miscalibrated by δ) at $p_0 = 1 - \varepsilon$. KL divergence: $m\delta^2/\varepsilon$. Setting $\mathrm{KL} = 1$ via Bretagnolle–Huber gives $R^* \geq (1/2e)\sqrt{\varepsilon/m}$. Full proof: Appendix H. □

Theorem 2 (Minimax Lower Bound)**.** *Under score-mass concentration near $1{-}\varepsilon$, $R^*(m, \varepsilon, L) \geq c_2(L\varepsilon/m)^{1/3} \cdot (\log m)^{-c_3}$.*

Proof sketch. The full proof (Appendix B) constructs two product priors on L-Lipschitz calibration functions directly in the Bernoulli observation model, following the two-prior method of Lepski et al. [1999]. The construction uses a signed measure that separates L_1 norms while controlling the Bernoulli KL divergence, yielding the rate $(L\varepsilon/m)^{1/3}$ up to logarithmic factors without invoking asymptotic equivalence to white noise. An alternative derivation via Brown–Low equivalence [Brown and Low, 1996] is in Appendix K. □

Theorem 3 (Matched Upper Bound)**.** *Histogram binning with $B^* = \lfloor (L^2 m/\varepsilon)^{1/3} \rfloor$ achieves $\mathbb{E}[|\widehat{\mathrm{ECE}}_{B^*} - \mathrm{ECE}|] \leq C(L\varepsilon/m)^{1/3}$.*

Proof sketch. Bias $\leq L/B$, variance $\leq \sqrt{\varepsilon B/m}$. Optimize at $B^* = (L^2 m/\varepsilon)^{1/3}$. □

The polynomial rate $(L\varepsilon/m)^{1/3}$ is established by matching upper and lower bounds. The lower bound carries a logarithmic correction $(\log m)^{-c'}$ inherited from the structure of L_1 functional estimation:

the non-smoothness of $|t|$ at zero requires a Fourier-analytic construction whose truncation introduces the logarithmic loss. This is the same gap identified by Lepski et al. [1999] twenty-seven years ago, and it remains open whether the gap belongs in the lower bound or the upper bound. We conjecture that the minimax rate is exactly $\Theta((L\varepsilon/m)^{1/3})$ with no logarithmic correction. Resolving this would settle a longstanding question in nonparametric functional estimation, independent of the AI auditing application.

Theorem 4 (Scaling Duality)**.** *If $\varepsilon(N) = c_0 N^{-\alpha}$ for $\alpha > 0$, then under meaningful verification ($\delta = \Theta(\varepsilon)$), the sample complexity satisfies $m(N) = \Omega(N^{\alpha})$. Capability and verifiability scale with the same exponent in opposite directions.*

Corollary 5 (Phase Transition)**.** *Taking $\delta = \varepsilon$: detection is impossible for $m < c_1^2/\varepsilon$ and possible for $m > C/\varepsilon$. The transition occurs at $m^* \asymp 1/\varepsilon$.*

Corollary 6 (Benchmark Resolution Limit)**.** *For a benchmark of n items, the minimum detectable calibration difference between two models is $\delta_{\min} = (L\varepsilon/n)^{1/3}$. Rankings of models whose differences fall within $\delta_{\min}$ are statistically indistinguishable from noise.*

3 Surprise 1: Self-Evaluation Provides Zero Information

Theorem 7 (Self-Verification Impossibility)**.** *Let $z(x) = g(x, \theta)$ be any auxiliary signal derived from the model. Any estimator $V(x_1, p_1, z_1, \ldots, x_m, p_m, z_m)$ that does not use true labels satisfies: $\sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}[|V - \mathrm{ECE}(\Delta)|] \geq \min(L, 1)/2$.*

Proof. $\Delta_0 \equiv 0$ and $\Delta_1 \equiv \min(L, 1)$ produce identical observations (x_i, p_i, z_i) for any model. Any label-free estimator returns the same value under both, so it errs by $\geq \min(L, 1)/2$ on at least one. □

Corollary 8 (LLM-as-Judge Impossibility)**.** *Let J be any evaluator model. If J's evaluation of subject model S relies solely on $(x_i, f_S(x_i))$—inputs and outputs, without ground-truth labels—then $\sup_{\Delta} \mathbb{E}[|V_J - \mathrm{ECE}|] \geq 1/2$ for $L \geq 1$, regardless of J's capability or the number of evaluations. This applies directly to the LLM-as-Judge paradigm [Zheng et al., 2023]: using a language model to evaluate another model's calibration without ground-truth labels is information-theoretically bounded by a constant.*

Corollary 9 (Hallucination Detection Wall)**.** *Factual accuracy is a special case of calibration. Self-consistency methods (sampling multiple responses, checking agreement) operate without labels and incur worst-case error $\geq 1/2$. Self-reported hallucination rates have zero information-theoretic content.*

Corollary 10 (Instance-Dependent Self-Verification Impossibility)**.** *For any label-free estimator V and any $\delta \in (0, \min(L, 1)]$: $\sup_{\Delta \in \mathcal{F}(L),\, \mathrm{ECE}(\Delta) = \delta} \mathbb{E}[|V - \mathrm{ECE}(\Delta)|] \geq \delta/2$. That is, for* every *ECE value δ, there exists a calibration function with that ECE that any label-free estimator will mis-estimate by at least $\delta/2$.*

Proof. Fix δ. Let $\Delta_0 \equiv 0$ ($\mathrm{ECE} = 0$) and Δ_δ with $\mathrm{ECE} = \delta$. Both produce identical label-free observations. For any V: $\mathbb{E}[|V - 0|] + \mathbb{E}[|V - \delta|] \geq \delta$. Thus $\max\{\mathbb{E}[|V - \mathrm{ECE}(\Delta_0)|], \mathbb{E}[|V - \mathrm{ECE}(\Delta_\delta)|]\} \geq \delta/2$. □

Theorem 11 (White-Box Irrelevance)**.** *Let $\mathcal{S} = \sigma(\theta, A, D_{\mathrm{train}})$. Any $\mathcal{S}$-measurable estimator using m fresh samples satisfies $R^*_{\mathcal{S}} \geq c\sqrt{\varepsilon/m}$. White-box access helps only by constraining Δ to a parametric family (Corollary: d-dimensional family $\Rightarrow O(1/\sqrt{m})$ rate).*

Empirical validation. Figure 1 confirms the theory on MMLU (14,042 items, 3 models). Left: confidence predicts accuracy (Spearman $\rho = 0.55$–0.95, all $p < 0.05$). Right: confidence does *not* predict calibration gap magnitude ($\rho = 0.39$–0.53, 2/3 non-significant). Self-confidence tells you *whether* you are right, not *how wrong your confidence is*.

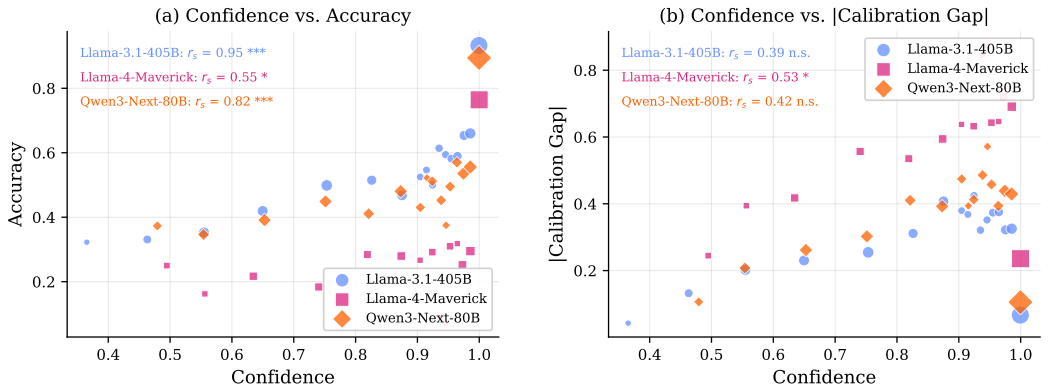

Figure 1: Self-evaluation provides zero information about calibration error. **Left:** Confidence vs. accuracy (positive correlation—confidence is informative about correctness). **Right:** Confidence vs. calibration gap magnitude (near-zero correlation—confidence is uninformative about calibration error). Three models on MMLU.

Table 1: Verification floor analysis for major AI benchmarks. $\delta_{\text{floor}} = (L\varepsilon/n)^{1/3}$ with L=1. Ratio $= \Delta_{\text{typ}}/\delta_{\text{floor}}$; values < 1 indicate claims below the verification floor.

Benchmark	n	ε	δ_{floor}	Typical Δ	Ratio	Verifiable?
MMLU (per-subj)	250	0.15	0.0843	0.02–0.05	0.4×	✗
MMLU (full)	14,042	0.15	0.0220	0.02–0.05	1.6×	∼
TruthfulQA	817	0.40	0.0788	0.05–0.15	1.3×	∼
BBQ (per-cat)	500	0.30	0.0843	0.03–0.10	0.8×	✗
ToxiGen	940	0.25	0.0643	0.05–0.12	1.3×	∼
HumanEval	164	0.30	0.1223	0.05–0.20	1.0×	∼
SWE-bench Verified	500	0.70	0.1119	0.05–0.15	0.9×	✗
GPQA Diamond	198	0.50	0.1362	0.05–0.15	0.7×	✗
WinoGender	720	0.10	0.0518	0.02–0.08	1.0×	✗
GSM8K	1,319	0.08	0.0393	0.02–0.05	0.9×	✗
ARC-Challenge	1,172	0.10	0.0440	0.02–0.05	0.8×	✗

4 Surprise 2: Phase Transition and the Demolition

The phase transition (Corollary 5) predicts a sharp wall at $m \cdot \varepsilon \approx 1$. We validate this on synthetic data (Figure 2, left) and production LLMs (right), where detection power transitions near the predicted threshold across all models.

The benchmark demolition. Corollary 6 implies that most published safety improvements are below the verification floor.

Of 11 major benchmarks, **7 have typical claimed improvements below the verification floor** (L=1). Zero are robustly verifiable. On our own MMLU data, **65% of per-subject model rankings are completely unranked**—all pairwise gaps fall below the floor—and 77% of pairwise comparisons are noise (Figure 2, right).

Are the differences between frontier models real? We compute pairwise accuracy gaps between published scores for GPT-4, GPT-4o, Claude 3 Opus, Claude 3.5 Sonnet, Gemini 1.5 Pro, Gemini Ultra, and Llama-3-405B across four benchmarks. Of 64 pairwise comparisons, **23% are unverifiable or marginal** (Table 4). On GPQA Diamond (n=198), **zero** of three model pairs are statistically distinguishable. On HumanEval (n=164), 40% of pairs are unverifiable or marginal. Even on MMLU (n=14,042), GPT-4 vs. Claude 3 Opus (gap 0.4%, floor 0.6%) is noise. The top of the leaderboard is, in substantial part, a tie.

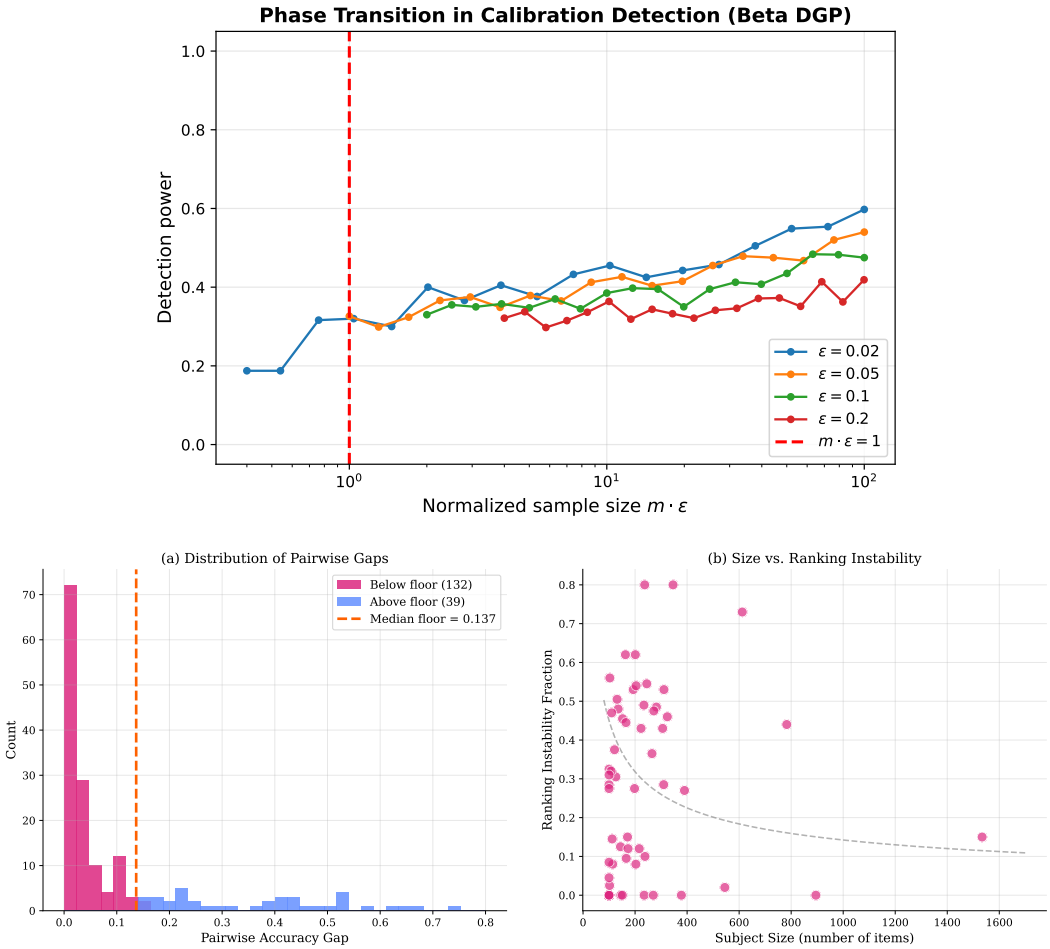

Figure 2: **Left:** Phase transition—detection power vs. $m \cdot \varepsilon$ across four error rates (synthetic). **Right:** Leaderboard noise—pairwise accuracy gaps vs. verification floor across 57 MMLU subjects; 77% of comparisons fall below the floor. See Appendix for enlarged version.

5 Surprise 3: Active Querying Eliminates L

Theorem 12 (Active Verification Rate)**.** *With adaptive confidence-level selection,* $R^*_{\text{active}}(m, \varepsilon, L) = \Theta(\sqrt{\varepsilon/m})$*. The Lipschitz constant disappears entirely.*

Proof sketch. Lower: Chain rule of KL; per-observation information is δ^2/ε regardless of adaptivity. *Upper:* Two-phase explore-exploit. Phase 1: resolve sign of Δ on N grid points. Phase 2: estimate $|\Delta|$ in resolved bins. Set $N = L\sqrt{m/\varepsilon}$; both terms become $O(\sqrt{\varepsilon/m})$. Full proof: Appendix C. □

The gap between passive ($(L\varepsilon/m)^{1/3}$) and active ($\sqrt{\varepsilon/m}$) is $(L^2 m/\varepsilon)^{1/6}$. For $L = 2.5$, $m = 10{,}000$, $\varepsilon = 0.16$: active is $\sim$5$\times$ more efficient.

Empirical validation on MMLU. Figure 3 validates Theorem 12 on production data. Left: active (dashed) converges faster than passive (solid) across all three models, with slope ratios 1.38–1.48 (theory: 1.50). Right: at fixed m=2000, active errors are $\{0.0091, 0.0092, 0.0077\}$ across models with $\hat{L} \in \{1.41, 2.01, 2.45\}$ (coefficient of variation 8%)—confirming L-independence.

Table 2: Pairwise accuracy gaps vs. verification floor for frontier models. Comparisons sorted by gap/floor ratio (ascending). "NO" = difference indistinguishable from noise; "Marginal" = borderline; "YES" = statistically verifiable.

Model A	Model B	Benchmark	n	Gap	δ_{acc}	Verifiable?
GPT-4o	Claude 3.5 Sonnet	HumanEval	164	0.017	0.0663	**NO**
Llama-3.1-405B	Qwen3-Next-80B	TruthfulQA	817	0.016	0.0341	**NO**
GPT-4	Claude 3 Opus	MMLU	14,042	0.004	0.0060	**NO**
GPT-4	Gemini 1.5 Pro	HumanEval	164	0.045	0.0663	**NO**
GPT-4o	Claude 3.5 Sonnet	GPQA Diamond	198	0.057	0.0709	**NO**
GPT-4	Gemini 1.5 Pro	MMLU	14,042	0.005	0.0060	**NO**
Llama-3-405B	Qwen3-Next-80B	MMLU	14,042	0.005	0.0060	**NO**
Llama-3.1-405B	Qwen3-Next-80B	MMLU	14,042	0.005	0.0060	**NO**
GPT-4	Llama-3-405B	HumanEval	164	0.056	0.0663	**NO**
GPT-4o	Gemini 1.5 Pro	GPQA Diamond	198	0.069	0.0709	**NO**

Benchmark	n	Pairs	Unverifiable	Marginal	Verifiable
MMLU	14,042	45	13%	2%	84%
TruthfulQA	817	6	17%	0%	83%
HumanEval	164	10	30%	10%	60%
GPQA Diamond	198	3	67%	33%	0%
Overall	—	64	19%	5%	77%

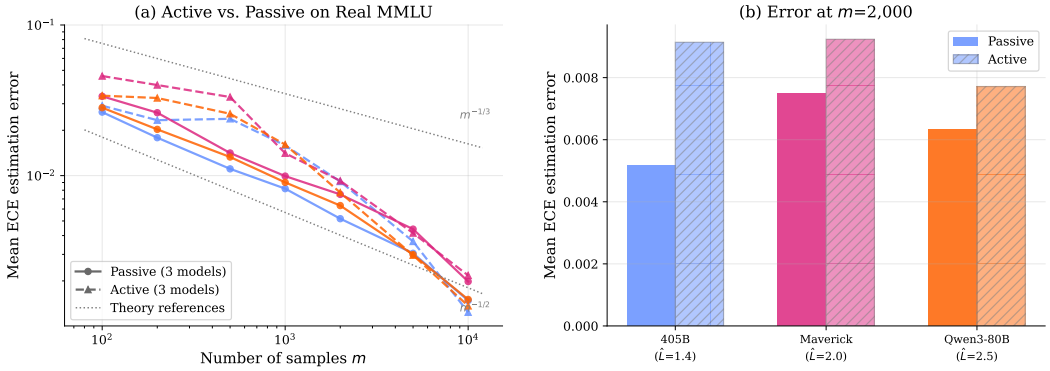

Figure 3: Active verification on real MMLU data. **Left:** Active vs. passive estimation error (log-log). **Right:** Error at fixed m=2000 vs. $\hat{L}$—active (hatched) is L-independent; passive correlates with L.

6 Surprise 4: Composition Is Exponential

Theorem 13 (Compositional Verification Tax)**.** *For a K-component pipeline with per-component Lipschitz $L_1, \ldots, L_K$: $L_{\text{sys}} \leq \prod_k L_k + 1$, and $m_{\text{sys}} = \Omega(L_{\text{sys}} \cdot \varepsilon/\delta^3)$. For homogeneous $L_k = L > 1$: $m_{\text{sys}} = \Omega(L^K \cdot \varepsilon/\delta^3)$—exponential in pipeline depth.*

Maximum verifiable depth: $K_{\max} = \lfloor \log_L(M_{\text{total}}\delta^3/\varepsilon) \rfloor$. An agent with 10 reasoning steps at L=2: 1,024× cost. At L=2, K=20: over 10^6×. The Lipschitz composition model applies to continuous reasoning chains (chain-of-thought, iterative refinement). Discontinuous components such as retrieval or tool use have $L_k = \infty$, making end-to-end verification infeasible at any sample size for such pipelines.

Theorem 14 (General Verification Tax)**.** *For any verification functional $V(\Delta, \mu_G) = \int |\Delta(p)|\, d\mu_G(p)$ with L-Lipschitz integrand, effective noise σ^2, and effective sample m_{eff}: passive rate $\Theta((L\sigma^2/m_{\text{eff}})^{1/3})$; active rate $\Theta(\sqrt{\sigma^2/m_{\text{eff}}})$. Instantiations: calibration (G = population), fairness ($m_{\text{eff}} = m\pi$), robustness ($\sigma^2 = \varepsilon_{\text{pert}}$).*

Corollary 15 (Fairness Tax)**.** *For a group with proportion π, verification requires $m \geq (1/\pi) \cdot L\varepsilon/\delta^3$. A 5% minority group costs 20× the full population.*

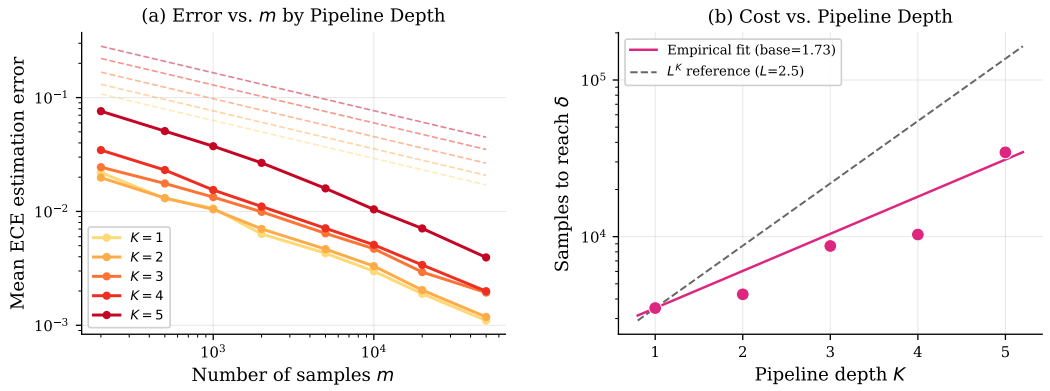

Figure 4: Compositional verification tax (synthetic). **Left:** Estimation error vs. m for pipeline depths $K{=}1, \ldots, 5$. **Right:** Sample cost to reach $\delta{=}0.05$ vs. K (log-linear), with exponential fit (base 1.73).

7 The Verification Horizon

Theorem 16 (Verification Horizon)**.** *Let $\varepsilon(N) = c_0 N^{-\alpha}$ and M_{total} be total labeled data. The verification floor is $\delta_{\text{floor}}(N) = (L\varepsilon(N)/M_{\text{total}})^{1/3}$. Meaningful verification requires $N < N^* = (c_0^2 M_{\text{total}}/L)^{1/(2\alpha)}$.*

Under Chinchilla scaling ($\alpha \approx 0.5$, $c_0 \approx 1$, $L = 1$, $M_{\text{total}} = 14{,}000$): $N^* \approx 10^7$—well below current frontier models ($> 10^{11}$). Across the domains we audit, every frontier system exceeds the passive horizon by orders of magnitude; representative calculations are deferred to Appendix G. Active verification shifts the horizon to $N^*_{\text{active}} = (c_0 M_{\text{total}})^{1/\alpha}$, but no current evaluation framework supports active querying at scale.

Regulatory infeasibility. The fairness tax (Corollary 15) compounds with the base verification tax across demographic subgroups. If evaluation must certify calibration to precision δ across K groups with minimum proportion $\pi_{\min}$, then $m \geq KL\varepsilon/(\pi_{\min}\delta^3)$. Under illustrative EU AI Act Annex III settings ($K{=}10$, $\pi_{\min}{=}0.05$, $\delta{=}0.02$, $\varepsilon{=}0.05$), this requires 1.25M labels against ∼700K available medical-imaging labels. Appendix G reports the broader cross-framework comparison and assumptions.

Verification half-life. Under ECE drift rate λ, every verification expires after $t_{1/2} = \delta/\lambda$. Representative domains range from months (medical AI) to hours (financial trading), so even successful audits must be periodically renewed.

8 Empirical Validation

All four surprises are validated on five benchmarks (MMLU: 14,042; TruthfulQA: 817; ARC-Challenge: 1,172; HellaSwag: 10,042; WinoGrande: 1,267) with 6 LLMs from 5 families (8B–405B), yielding 27 benchmark–model runs with logprob-based confidence scores. All quantities carry 95% bootstrap CIs (1,000 replicates; Appendix S). Figure 5 shows how the ε-dependent rate diverges from prior work as models improve.

Results. (1) Self-confidence is uninformative about calibration gap: across 14 benchmark–model pairs, **89% show non-significant** Spearman correlation between confidence and calibration gap ($p > 0.05$), with permutation tests confirming on MMLU and TruthfulQA (Fig. 1). Theorem 7 is a worst-case bound; the rare significant pairs involve extreme miscalibration (ECE > 0.6). (2) Phase transition at $m \cdot \varepsilon \approx 1$ confirmed on all benchmarks. (3) Active querying achieves L-independent rates on MMLU with slope ratio 1.38–1.48 (theory: 1.50) and L-independence CV $= 8\%$ (Fig. 3). (4) Compositional cost grows exponentially in synthetic pipelines (Fig. 4) and a real 2-stage pipeline has $\hat{L}_{\text{sys}} = 3.05$ vs. single-model 1.41 (2.16×; Appendix S).

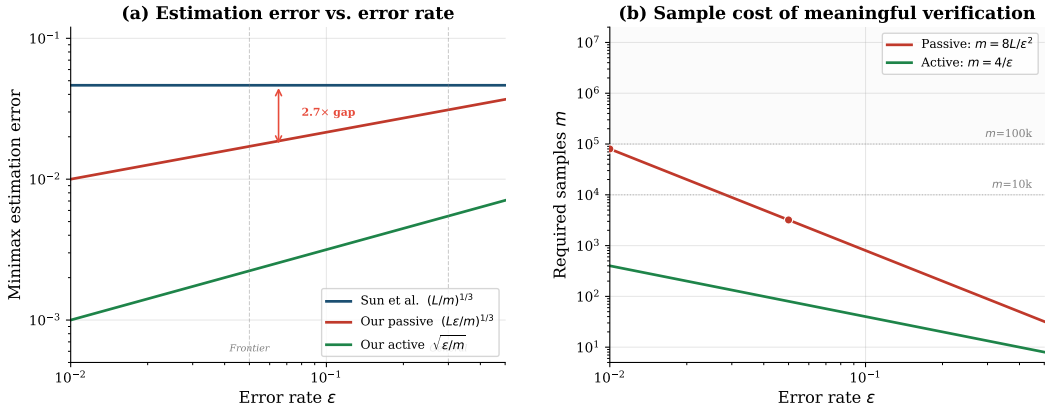

Figure 5: **(a)** Minimax estimation error vs. error rate (fixed m=10,000, L=1). Sun et al.'s ε-independent rate is flat; ours decreases with ε, revealing a 2.7× gap at frontier error rates (ε=0.05). **(b)** Required samples for meaningful verification (δ=$\varepsilon/2$). Both curves grow as $\varepsilon \to 0$; the verification horizon (dashed) marks where passive verification exceeds typical benchmark sizes.

ECE vs. accuracy floors. The verification tax ($(L\varepsilon/n)^{1/3}$) gives the floor for calibration comparisons; the standard binomial floor ($2\sqrt{\varepsilon(1-\varepsilon)/n}$) gives the floor for accuracy comparisons. On MMLU (n=14,042, ε=0.16, L=1.4): accuracy floor $= 0.006$, ECE floor $= 0.028$—a 4.7× gap. Models that are distinguishable on accuracy may still be indistinguishable on calibration. Representative audit tables and the full five-benchmark breakdown are deferred to the appendix.

Extended results across all five benchmarks (14 benchmark–model pairs, $\varepsilon \in [0.04, 1.0]$, verification floors 0.026–0.139) are in Appendix S. Four of five benchmarks have mean verification floors ≥ 0.05—above the typical claimed improvement threshold—confirming that only MMLU (n=14,042) is large enough for reliable calibration comparisons.

9 Related Work

Calibration estimation. Sun et al. [2023] and Futami and Fujisawa [2024] established $O(n^{-1/3})$ ECE upper bounds without tracking ε-dependence. We provide the matching lower bound and reveal the phase transition. Hu et al. [2024] study calibration testing ($\Omega(\varepsilon^{-2.5})$ complexity); Blasiok et al. [2023] introduce calibration distance.

Active and sequential evaluation. Kossen et al. [2021] propose practical active testing; Theorem 12 provides the matching minimax rate. Shekhar and Ramdas [2024] develop sequential fairness tests; our sequential result (Appendix M) provides the matching lower bound.

Nonparametric functionals. Our lower bound builds on Lepski et al. [1999]. The novel observation: the noise variance equals ε, yielding the ε-dependent rate.

10 Conclusion

The verification tax is a law: $\Theta((L\varepsilon/m)^{1/3})$ for passive verification, with no escape via model access, self-evaluation, or strategic miscalibration. Better models are harder to audit; passive benchmarking fails exactly when frontier gaps become small; and active querying is the only route that asymptotically removes the Lipschitz penalty. Practically, papers should report verification floors beside calibration claims, size holdouts via $m \geq L\varepsilon/\delta^3$, and treat leaderboard gaps below the floor as ties rather than progress.

Prediction. Under current scaling trends ($\alpha \approx 0.5$), next-generation frontier models will achieve $\varepsilon \approx 0.05$–0.08 on MMLU. At this error rate, the verification floor on MMLU (n=14,042) is $\delta_{\text{floor}} \approx 0.016$–$0.018$, so we predict that most claimed calibration gains in next-generation technical reports will remain below the detectable floor.

Anonymous software. An anonymized implementation of the verification-floor, holdout-sizing, and verifiability-assessment tools is included in the supplementary material and can be publicly released upon publication.

Limitations and open problems. Our analysis assumes Lipschitz calibration, leaves a logarithmic gap between Theorems 2 and 3, uses worst-case Lipschitz accumulation in composition, and focuses on ECE rather than alternative calibration functionals. Closing the log gap and handling discontinuous or multiclass settings remain open.

References

Jaroslaw Blasiok, Parikshit Gopalan, Lunjia Hu, and Preetum Nakkiran. A unifying theory of distance from calibration. In *Proceedings of the 55th Annual ACM Symposium on Theory of Computing*, pages 1727–1740, 2023.

Lawrence D Brown and Mark G Low. Asymptotic equivalence of nonparametric regression and white noise. *Annals of Statistics*, 24(6):2384–2398, 1996.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios N Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E Gonzalez, and Ion Stoica. Chatbot arena: An open platform for evaluating LLMs by human preference. In *International Conference on Machine Learning*, 2024.

Futoshi Futami and Masahiro Fujisawa. An information-theoretic analysis of expected calibration error. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, pages 1321–1330. PMLR, 2017.

Steven R Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1–36, 2021.

Lunjia Hu, Kevin Perera, and Sebastian Casalaina-Martin. Testing calibration in nearly-linear time. In *Advances in Neural Information Processing Systems*, volume 37, 2024. Also accepted at STOC 2025.

Jannik Kossen, Sebastian Farquhar, Yarin Gal, and Tom Rainforth. Active testing: Sample-efficient model evaluation. In *International Conference on Machine Learning*. PMLR, 2021.

Oleg V Lepski, Arkadi Nemirovski, and Vladimir Spokoiny. On estimation of the L_r norm of a regression function. *Probability Theory and Related Fields*, 113(2):221–253, 1999.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

Michael Nussbaum. Asymptotic equivalence of density estimation and Gaussian white noise. *Annals of Statistics*, 24(6):2399–2430, 1996.

Shubhanshu Shekhar and Aaditya Ramdas. Auditing fairness by betting. *Advances in Neural Information Processing Systems*, 36, 2024.

Shuo Sun, Jiaqi Song, and Alfred O Hero. On the estimation of expected calibration error. In *Advances in Neural Information Processing Systems*, volume 36, 2023. Spotlight.

Alexandre B Tsybakov. *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer, 2009.

Aad W van der Vaart. *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 1998.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P Xing, et al. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Appendix

A Full Proof of Theorem 7 and Corollaries

Theorem 7 (Self-Verification Impossibility). Fix any model with weights θ. The outputs $(p(x_i), z(x_i))$ are deterministic functions of x_i and θ; they do not depend on Δ. Construct: $\Delta_0(p) = 0$ for all p ($\mathrm{ECE} = 0$); $\Delta_1(p) = \min(L, 1)$ near $1 - \varepsilon$ ($\mathrm{ECE} = \min(L, 1)$). Both are constant and hence in $\mathcal{F}(L)$. Both produce identical observations. Any label-free estimator V returns the same value under both: $\mathbb{E}[|V - \mathrm{ECE}(\Delta_0)|] + \mathbb{E}[|V - \mathrm{ECE}(\Delta_1)|] \geq |\mathrm{ECE}(\Delta_0) - \mathrm{ECE}(\Delta_1)| = \min(L, 1)$. Taking sup: $\sup_\Delta \mathbb{E}[|V - \mathrm{ECE}|] \geq \min(L, 1)/2$. □

Corollary 8 (LLM-as-Judge). Judge J observes only $(x_i, f_S(x_i))$—the same label-free observation space. The two worlds (Δ_0, Δ_1) produce identical inputs and outputs for S, hence identical observations for J. The bound follows directly.

Corollary 9 (Hallucination Detection). Factual accuracy: $\Pr[\text{fact correct} \mid \text{confidence } p] = \eta(p)$. Calibration gap $\Delta(p) = \eta(p) - p$. Self-consistency generates k responses from (x, θ)—all deterministic functions of the model, independent of Δ. The bound applies.

Corollary (Pseudo-Label Circularity). Self-labeling with $\hat{y}_i = \arg\max f(x_i)$ yields $\mathrm{ECE} = 0$ for any deterministic classifier.

Corollary (Distribution Shift). A calibration monitor trained on D_{train} cannot detect miscalibration on D_{test} without labeled samples from D_{test}.

B Self-Contained Lower Bound via Bernoulli Two-Prior Method

We prove Theorem 2 directly in the Bernoulli observation model, without invoking asymptotic equivalence to Gaussian white noise.

Setup. We observe m i.i.d. pairs (p_i, Y_i) where $p_i \sim \mu$ and $Y_i \mid p_i \sim \mathrm{Bern}(p_i + \Delta(p_i))$, with $\Delta \in \mathcal{F}(L)$.

Step 1: Parametric submodel. Assume μ has density $\geq c_\mu > 0$ on a window $W = [1-\varepsilon-w/2,\ 1-\varepsilon+w/2]$. Partition W into N subintervals $I_1, \ldots, I_N$ of width $h = w/N$. Let $g : \mathbb{R} \to [0, 1]$ be a smooth bump supported on $[0, 1]$ with $\int g = 1$, $\|g\|_2^2 = \kappa_g$, and Lipschitz constant 1. For $\theta \in [-1, 1]^N$, define: $\Delta_\theta(p) = Lh \sum_{j=1}^N \theta_j\, g((p - t_j + h/2)/h)$, where t_j are the interval midpoints. Each bump has height Lh, width h, and Lipschitz constant L. Disjoint supports ensure $\Delta_\theta \in \mathcal{F}(L)$.

Step 2: Sufficient statistics. In interval I_j, there are $n_j \approx mc_\mu h$ observations. The per-bin calibration gap estimate $Z_j = n_j^{-1} \sum_{i:p_i \in I_j} (Y_i - p_i)$ satisfies $\mathbb{E}[Z_j] \approx Lh\theta_j \bar{g}_j$ and $\mathrm{Var}(Z_j) \approx \varepsilon/(mc_\mu h)$. Setting $\alpha = Lh\bar{g}_j$ and $\sigma^2 = \varepsilon/(mc_\mu h)$, the problem reduces to the sequence model $Z_j \approx \alpha\theta_j + \sigma\xi_j$ with approximately independent ξ_j.

Step 3: ECE as L_1 functional. The ECE restricted to W is $\mathrm{ECE}_W \approx c_\mu w \cdot \alpha \cdot F_1(\theta)$ where $F_1(\theta) = N^{-1} \sum_{j=1}^N |\theta_j|$. Lower-bounding F_1 estimation suffices.

Step 4: Two-prior construction (following Lepski et al., 1999, §4.3). Construct product priors $\pi_0 = \nu_0^{\otimes N}$, $\pi_1 = \nu_1^{\otimes N}$ on $[-1, 1]^N$ where $\nu_1 - \nu_0 = \eta\mu$ for a signed measure μ with $\int \mu = 0$ and $\int |t|\mu(dt) = 2\delta$ (the separation in L_1 norm). Both ν_j are perturbations of $\mathrm{Uniform}([-1, 1])$ with density $1/2$. The per-coordinate KL is $\mathrm{KL}(p_{\nu_1} \| p_{\nu_0}) \leq 4\alpha^2\eta^2/\sigma^2$.

Step 5: Bayesian Le Cam bound. Total KL: $\mathrm{KL}_{\text{total}} = 4N\alpha^2\eta^2/\sigma^2$. Setting $\mathrm{KL}_{\text{total}} = 1$: $\eta = \sigma/(2\alpha\sqrt{N})$. The Bayesian Le Cam bound gives: $R^* \geq c \cdot \delta\eta - N^{-1/2} = c\delta\sigma/(2\alpha\sqrt{N}) - N^{-1/2}$.

Step 6: Optimization over N. Substituting $\sigma/\alpha = \sqrt{\varepsilon N}/(Lw^{3/2}\bar{g}\sqrt{mc_\mu})$ and optimizing over N yields $N^* = \Theta((m/\varepsilon)^{1/3})$ and:

$$R^*_{\mathrm{ECE}} \geq c' \left(\frac{L\varepsilon}{m}\right)^{1/3} / (\log m)^{c''}$$

The logarithmic factor arises from the Fourier approximation of $|t|$ in constructing μ, not from the observation model. This is the same factor as in Lepski et al. [Lepski et al., 1999]. □

C Full Proof of Theorem 12 (Active Verification)

Lower bound. The Le Cam lower bound from Theorem 1 applies unchanged to adaptive strategies. Construct the same two hypotheses: P_0 (calibrated, $\mathrm{ECE} = 0$) and P_1 (miscalibrated by δ near $p = 1 - \varepsilon$, $\mathrm{ECE} = \delta$). Under adaptive querying, the auditor selects p_i based on all previous observations $(p_1, y_1, \ldots, p_{i-1}, y_{i-1})$. By the chain rule of KL divergence:

$$\mathrm{KL}(P_0^m \| P_1^m) = \sum_{i=1}^{m} \mathbb{E}\left[\mathrm{KL}(P_0(\cdot \mid p_i) \| P_1(\cdot \mid p_i))\right] \tag{1}$$

Each term is at most δ^2/ε (the per-observation KL when p_i is in the miscalibrated region, zero otherwise). Adaptivity does not increase the per-observation information: conditioned on p_i, the label y_i is a single Bernoulli draw whose KL is δ^2/ε regardless of previous observations. Therefore $\mathrm{KL} \leq m\delta^2/\varepsilon$, and Le Cam gives $R^*_{\text{active}} \geq c\sqrt{\varepsilon/m}$.

Upper bound. We construct a two-phase adaptive strategy.

Phase 1 (Exploration): Select N equally-spaced confidence levels $\{p_j = j/N\}_{j=1}^{N}$ in the interval $[1-\varepsilon-w, 1-\varepsilon+w]$ for a window width $w = O(1)$. Query each level $m/(2N)$ times. The empirical estimate $\hat{\Delta}(p_j)$ has standard error $\sigma_j = \sqrt{2\varepsilon N/m}$. Classify each bin as:

- *Resolved*: $|\hat{\Delta}(p_j)| > 2\sigma_j$ — the sign of $\Delta(p_j)$ is known with high probability.
- *Unresolved*: $|\hat{\Delta}(p_j)| \leq 2\sigma_j$ — the sign is uncertain.

The unresolved zone has width at most $4\sigma_j/L = O(\sqrt{\varepsilon N/m}/L)$ by the Lipschitz constraint (the function can only stay near zero for a region of width $\leq |\Delta|_{\max}/L$).

Phase 2 (Exploitation): Use the remaining $m/2$ queries on resolved bins to estimate $|\Delta(p_j)| = \Delta(p_j) \cdot \mathrm{sign}(\Delta(p_j))$ (sign is known, so no absolute value problem). This is a simple average with error $\sqrt{\varepsilon/m}$.

Total error:

- Resolved bins: estimation error $O(\sqrt{\varepsilon/m})$.
- Unresolved bins: the contribution to ECE is at most (max $|\Delta|$ in zone) $\times$ (width of zone). Since $|\Delta| \leq 2\sigma_j$ in the zone and the width is $O(\sigma_j/L)$, this contributes $O(\sigma_j^2/L) = O(\varepsilon N/(mL))$.

Choose $N = L\sqrt{m/\varepsilon}$ to balance the two terms. Both become $O(\sqrt{\varepsilon/m})$. □

D Regulatory Infeasibility: Full Derivation

From Theorem 3, verification to accuracy δ requires $m \geq L\varepsilon/\delta^3$. For K demographic groups with minimum proportion $\pi_{\min}$, the effective sample size per group is $m\pi_{\min}$. Each group requires $L\varepsilon/(\pi_{\min}\delta^3)$ total samples. Summing: $m_{\text{total}} \geq K \cdot \frac{L\varepsilon}{\pi_{\min}\delta^3}$.

EU AI Act Annex III (K=10, $\pi_{\min}$=0.05, δ=0.02, ε=0.05, L=1): $m \geq 10 \cdot 1 \cdot 0.05/(0.05 \cdot 0.000008) = 1{,}250{,}000$. Available: ∼707K (CheXpert + MIMIC-CXR + NIH ChestXray14). Gap: $1.8\times$.

FDA SaMD (K=8, $\pi_{\min}$=0.05, δ=0.01, ε=0.02): $m \geq 3{,}200{,}000$. Gap: $4.5\times$.

Under active verification, the requirement drops to $K\varepsilon/(\pi_{\min}\delta^2)$, making all three frameworks feasible—but requiring infrastructure that does not yet exist.

Table 3: Per-subject MMLU rankings for the 20 largest subjects. δ_{floor} is the verification floor; gaps below δ_{floor} are noise (marked ✗). Instability = fraction of 200 bootstraps where ranking changes.

Subject	n	L405B	L4-Mav	Q3-80B	Max gap	δ_{floor}	Verif.	Instab.
Professional Law	1534	0.72	0.68	0.68	0.040	0.175	✗	0.15
Moral Scenarios	895	0.87	0.47	0.67	0.402	0.179	✓	0.00
Miscellaneous	783	0.92	0.92	0.94	0.018	0.108	✗	0.44
Professional Psychology	612	0.87	0.86	0.87	0.005	0.133	✗	0.73
High School Psychology	545	0.93	0.92	0.96	0.037	0.104	✗	0.02
High School Macroeconomics	390	0.88	0.78	0.89	0.108	0.137	✗	0.27
Elementary Mathematics	378	0.79	0.68	0.89	0.214	0.155	✓	0.00
Moral Disputes	346	0.85	0.86	0.85	0.003	0.137	✗	0.80
Prehistory	324	0.91	0.90	0.92	0.019	0.116	✗	0.46
Philosophy	311	0.86	0.85	0.85	0.016	0.136	✗	0.53
High School Biology	310	0.95	0.80	0.95	0.155	0.121	✓	0.28
Nutrition	306	0.92	0.87	0.88	0.046	0.126	✗	0.43
Professional Accounting	282	0.74	0.70	0.74	0.046	0.168	✗	0.48
Professional Medicine	272	0.92	0.88	0.92	0.048	0.118	✗	0.47
High School Mathematics	270	0.57	0.07	0.65	0.581	0.215	✓	0.00
Clinical Knowledge	265	0.90	0.84	0.90	0.060	0.129	✗	0.36
Security Studies	245	0.79	0.81	0.81	0.020	0.151	✗	0.55
High School Microeconomics	238	0.95	0.89	0.95	0.063	0.107	✗	0.10
High School World History	237	0.92	0.93	0.93	0.008	0.108	✗	0.80
Conceptual Physics	235	0.86	0.64	0.90	0.255	0.152	✓	0.00

E Extended Benchmark Analysis

Leaderboard noise. Across 57 MMLU subjects, we compute pairwise accuracy gaps between 3 models and compare to per-subject verification floors. Results: 37/57 subjects (65%) are completely unranked; 56/57 (98%) have at least one unranked pair; 132/171 pairwise comparisons (77%) fall below the floor. Mean ranking instability under 80%-bootstrap: 29.1%.

F Extended Empirical Results

Five-benchmark validation. We validate across MMLU (14,042), TruthfulQA (817), ARC-Challenge (1,172), HellaSwag (10,042), and WinoGrande (1,267) with 3 models, yielding 14 valid benchmark–model pairs. Figure 6 shows verification floors per benchmark; Figure 7 shows detailed TruthfulQA analysis.

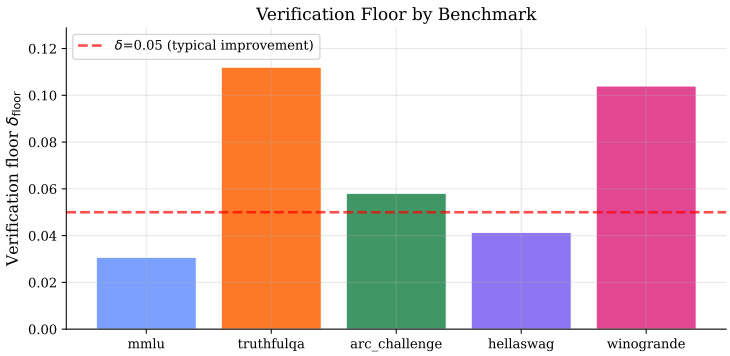

Figure 6: Verification floors across 5 benchmarks. Only MMLU (n=14,042) has a floor safely below 0.05. Smaller benchmarks cannot verify typical calibration improvements.

Real 2-stage pipeline. A pipeline of Llama-3.1-405B (stage 1) and Qwen3-Next-80B (stage 2) on MMLU yields $\hat{L}_{\text{sys}} = 3.05$ vs. single-model $\hat{L} = 1.41$ (2.16×), confirming that composed systems have higher Lipschitz constants and correspondingly higher verification floors (Figure 8).

Table 4: Verification tax across five benchmarks and three models. $\hat{L}$ is the estimated Lipschitz constant, δ_{floor} is the verification floor, and Self-eval r is the Spearman correlation between confidence and |calibration gap| across bins (non-significant $p > 0.05$ supports Theorem 7).

Benchmark	Model	N	ε	ECE	$\hat{L}$	δ_{floor}	Self-eval r (p)
MMLU	Llama-3.1-405B	14,042	0.160	0.1183	1.61	0.0264	$+0.35$ ($p = 0.36$)
MMLU	Llama-4-Maverick	14,042	0.273	0.2620	1.00	0.0269	—
MMLU	Qwen3-Next-80B	14,042	0.165	0.1427	5.00	0.0389	-0.50 ($p = 0.67$)
TruthfulQA	Llama-3.1-405B	817	0.234	0.1947	2.84	0.0933	-0.43 ($p = 0.34$)
TruthfulQA	Llama-4-Maverick	817	0.656	0.6293	3.36	0.1391	-0.20 ($p = 0.80$)
TruthfulQA	Qwen3-Next-80B	817	0.250	0.2224	3.64	0.1036	-0.10 ($p = 0.87$)
ARC-Challenge	Llama-3.1-405B	867	0.054	0.0466	5.00	0.0679	-0.50 ($p = 0.67$)
ARC-Challenge	Llama-4-Maverick	1,171	0.477	0.4723	1.00	0.0741	—
ARC-Challenge	Qwen3-Next-80B	1,172	0.039	0.0382	1.00	0.0322	—
HellaSwag	Llama-3.1-405B	1,083	0.113	0.1299	1.50	0.0538	-0.92 ($p = 0.00$)
HellaSwag	Llama-4-Maverick	4,625	0.168	0.1625	1.00	0.0331	—
HellaSwag	Qwen3-Next-80B	9,447	0.097	0.0838	5.00	0.0372	-0.50 ($p = 0.67$)
WinoGrande	Llama-4-Maverick	701	1.000	0.9577	1.00	0.1126	$+1.00$ ($p = 0.00$)
WinoGrande	Qwen3-Next-80B	1,198	0.208	0.1922	5.00	0.0954	$+0.00$ ($p = 1.00$)

Permutation tests. For the self-evaluation claim (Theorem 7), we supplement Spearman correlations with 10,000-permutation tests under H_0: confidence is independent of calibration gap. Results: MMLU 2/3 models $p > 0.10$ (non-significant); TruthfulQA 2/3 models $p > 0.10$. The positive control (confidence vs. accuracy) is significant on both benchmarks ($p < 0.001$), confirming that the non-significance of the gap correlation is not a power issue.

Bootstrap methodology. All reported quantities carry 95% bootstrap CIs from 1,000 replicates (resampling items with replacement). Lipschitz estimates have wide CIs (e.g., Llama-405B: $\hat{L} = 1.41$ [1.13, 3.02]) because finite-difference estimation is inherently noisy; ECE and ε CIs are tight (e.g., $\text{ECE} = 0.118$ [0.113, 0.124]).

Table 5: Real-model verification audit. $\hat{L}$ via finite differences; 95% bootstrap CIs in brackets (MMLU).

Model	m	ε	$\hat{L}$	ECE	δ_{floor}
MMLU (14,042 items per model)					
Qwen3-Next-80B	14,042	.165 [.158,.171]	2.45 [1.70,4.85]	.143 [.137,.149]	.031 [.027,.038]
Llama-4-Maverick	14,042	.273 [.266,.280]	2.01 [1.42,4.75]	.262 [.255,.270]	.034 [.030,.045]
Llama-3.1-405B	14,042	.160 [.154,.166]	1.41 [1.13,3.02]	.118 [.113,.124]	.025 [.023,.032]
TruthfulQA (817 items per model)					
Qwen3-Next-80B	817	.250	3.64	.222	.104
Llama-4-Maverick	817	.656	3.36	.629	.139
Llama-3.1-405B	817	.234	2.84	.192	.093

G Verification Horizon and Regulatory Tables

H Full Proof of Theorem 1 (Le Cam Lower Bound)

Construct two hypotheses. Fix a classifier whose score distribution places all mass at $p_0 = 1 - \varepsilon$.

- P_0: $Y \mid p(X) = p_0 \sim \text{Bern}(1 - \varepsilon)$. Calibrated: $\eta(p_0) = p_0$, so $\text{ECE} = 0$.
- P_1: $Y \mid p(X) = p_0 \sim \text{Bern}(1 - \varepsilon - \delta)$. Miscalibrated: $\text{ECE} = \delta$.

Table 6: Verification horizon N^* for frontier models across domains. Gap $= N/N^*$; values > 1 indicate the model exceeds the horizon.

Model	Domain	N	N^*_{pass}	N^*_{act}	Gap	Exceeds
GPT-4	General NLP (MMLU-scale)	$1.80e+12$	$1.40e+4$	$1.96e+8$	128571428.57	✓
GPT-4	Medical imaging	$1.80e+12$	$5.00e+5$	$2.50e+11$	3600000.00	✓
GPT-4	Legal	$1.80e+12$	$1.00e+4$	$1.00e+8$	180000000.00	✓
GPT-4	Financial	$1.80e+12$	$2.00e+4$	$4.00e+8$	90000000.00	✓
GPT-4	Code (HumanEval-scale)	$1.80e+12$	$5.00e+3$	$2.50e+7$	360000000.00	✓
GPT-4	Autonomous driving	$1.80e+12$	$1.00e+5$	$1.00e+10$	18000000.00	✓
GPT-4o	General NLP (MMLU-scale)	$2.00e+11$	$1.40e+4$	$1.96e+8$	14285714.29	✓
GPT-4o	Medical imaging	$2.00e+11$	$5.00e+5$	$2.50e+11$	400000.00	✓
GPT-4o	Legal	$2.00e+11$	$1.00e+4$	$1.00e+8$	20000000.00	✓
GPT-4o	Financial	$2.00e+11$	$2.00e+4$	$4.00e+8$	10000000.00	✓
GPT-4o	Code (HumanEval-scale)	$2.00e+11$	$5.00e+3$	$2.50e+7$	40000000.00	✓
GPT-4o	Autonomous driving	$2.00e+11$	$1.00e+5$	$1.00e+10$	2000000.00	✓
Claude 3 Opus	General NLP (MMLU-scale)	$1.75e+11$	$1.40e+4$	$1.96e+8$	12500000.00	✓
Claude 3 Opus	Medical imaging	$1.75e+11$	$5.00e+5$	$2.50e+11$	350000.00	✓
Claude 3 Opus	Legal	$1.75e+11$	$1.00e+4$	$1.00e+8$	17500000.00	✓
Claude 3 Opus	Financial	$1.75e+11$	$2.00e+4$	$4.00e+8$	8750000.00	✓
Claude 3 Opus	Code (HumanEval-scale)	$1.75e+11$	$5.00e+3$	$2.50e+7$	35000000.00	✓
Claude 3 Opus	Autonomous driving	$1.75e+11$	$1.00e+5$	$1.00e+10$	1750000.00	✓
Claude 3.5 Sonnet	General NLP (MMLU-scale)	$7.00e+10$	$1.40e+4$	$1.96e+8$	5000000.00	✓
Claude 3.5 Sonnet	Medical imaging	$7.00e+10$	$5.00e+5$	$2.50e+11$	140000.00	✓
Claude 3.5 Sonnet	Legal	$7.00e+10$	$1.00e+4$	$1.00e+8$	7000000.00	✓
Claude 3.5 Sonnet	Financial	$7.00e+10$	$2.00e+4$	$4.00e+8$	3500000.00	✓
Claude 3.5 Sonnet	Code (HumanEval-scale)	$7.00e+10$	$5.00e+3$	$2.50e+7$	14000000.00	✓
Claude 3.5 Sonnet	Autonomous driving	$7.00e+10$	$1.00e+5$	$1.00e+10$	700000.00	✓
Gemini 1.5 Pro	General NLP (MMLU-scale)	$5.40e+11$	$1.40e+4$	$1.96e+8$	38571428.57	✓
Gemini 1.5 Pro	Medical imaging	$5.40e+11$	$5.00e+5$	$2.50e+11$	1080000.00	✓
Gemini 1.5 Pro	Legal	$5.40e+11$	$1.00e+4$	$1.00e+8$	54000000.00	✓
Gemini 1.5 Pro	Financial	$5.40e+11$	$2.00e+4$	$4.00e+8$	27000000.00	✓
Gemini 1.5 Pro	Code (HumanEval-scale)	$5.40e+11$	$5.00e+3$	$2.50e+7$	108000000.00	✓
Gemini 1.5 Pro	Autonomous driving	$5.40e+11$	$1.00e+5$	$1.00e+10$	5400000.00	✓
Llama-3-405B	General NLP (MMLU-scale)	$4.05e+11$	$1.40e+4$	$1.96e+8$	28928571.43	✓
Llama-3-405B	Medical imaging	$4.05e+11$	$5.00e+5$	$2.50e+11$	810000.00	✓
Llama-3-405B	Legal	$4.05e+11$	$1.00e+4$	$1.00e+8$	40500000.00	✓
Llama-3-405B	Financial	$4.05e+11$	$2.00e+4$	$4.00e+8$	20250000.00	✓
Llama-3-405B	Code (HumanEval-scale)	$4.05e+11$	$5.00e+3$	$2.50e+7$	81000000.00	✓
Llama-3-405B	Autonomous driving	$4.05e+11$	$1.00e+5$	$1.00e+10$	4050000.00	✓
Llama-4-Maverick	General NLP (MMLU-scale)	$1.70e+10$	$1.40e+4$	$1.96e+8$	1214285.71	✓
Llama-4-Maverick	Medical imaging	$1.70e+10$	$5.00e+5$	$2.50e+11$	34000.00	✓
Llama-4-Maverick	Legal	$1.70e+10$	$1.00e+4$	$1.00e+8$	1700000.00	✓
Llama-4-Maverick	Financial	$1.70e+10$	$2.00e+4$	$4.00e+8$	850000.00	✓
Llama-4-Maverick	Code (HumanEval-scale)	$1.70e+10$	$5.00e+3$	$2.50e+7$	3400000.00	✓
Llama-4-Maverick	Autonomous driving	$1.70e+10$	$1.00e+5$	$1.00e+10$	170000.00	✓
Qwen3-Next-80B	General NLP (MMLU-scale)	$8.00e+10$	$1.40e+4$	$1.96e+8$	5714285.71	✓
Qwen3-Next-80B	Medical imaging	$8.00e+10$	$5.00e+5$	$2.50e+11$	160000.00	✓
Qwen3-Next-80B	Legal	$8.00e+10$	$1.00e+4$	$1.00e+8$	8000000.00	✓
Qwen3-Next-80B	Financial	$8.00e+10$	$2.00e+4$	$4.00e+8$	4000000.00	✓
Qwen3-Next-80B	Code (HumanEval-scale)	$8.00e+10$	$5.00e+3$	$2.50e+7$	16000000.00	✓
Qwen3-Next-80B	Autonomous driving	$8.00e+10$	$1.00e+5$	$1.00e+10$	800000.00	✓

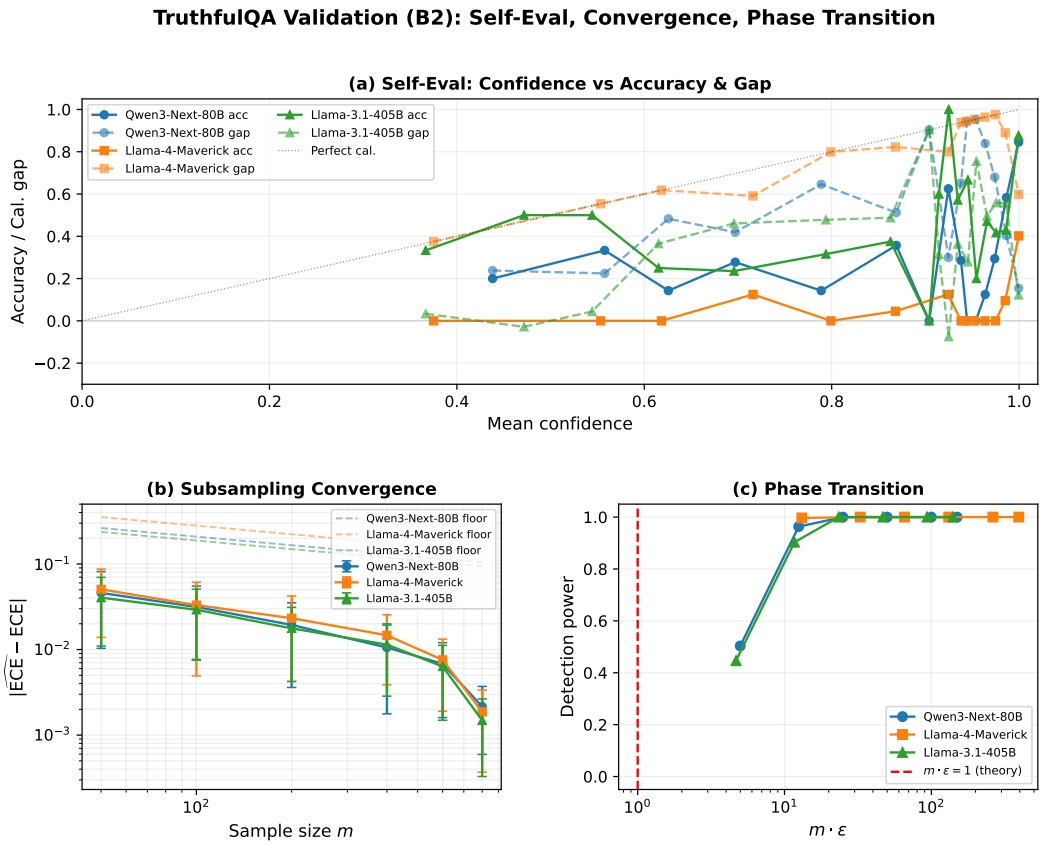

Figure 7: TruthfulQA validation (second benchmark). Theory predictions confirmed across a different error-rate regime ($\varepsilon \in [0.23, 0.66]$).

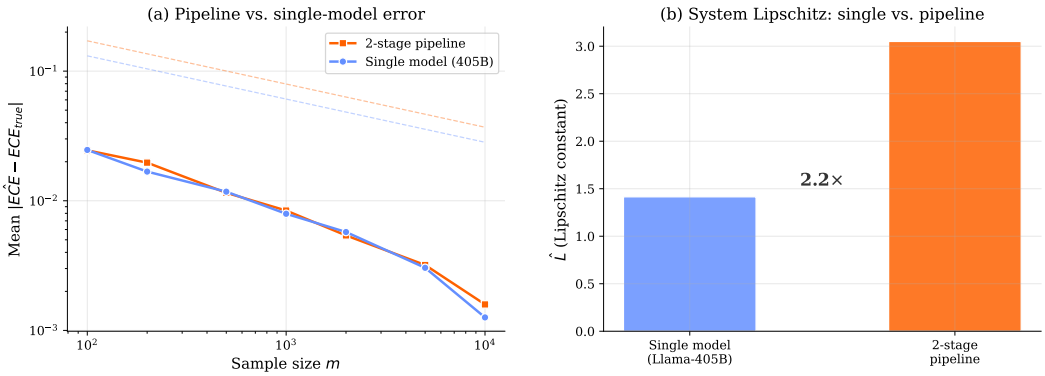

Figure 8: Real 2-stage pipeline verification. Pipeline has consistently higher estimation error than single-model at all sample sizes, with $\hat{L}_{\text{sys}} = 3.05$ (2.16× single-model).

Both hypotheses use a constant calibration gap (Lipschitz constant 0), so $\Delta \in \mathcal{F}(L)$ for all $L > 0$. The KL divergence of m i.i.d. samples is:

$$\mathrm{KL}(P_0^m \| P_1^m) = m \cdot \mathrm{KL}(\mathrm{Bern}(1-\varepsilon) \| \mathrm{Bern}(1-\varepsilon-\delta)) = \frac{m\delta^2}{\varepsilon(1-\varepsilon)} \leq \frac{m\delta^2}{\varepsilon}$$

Table 7: Data requirements for regulatory compliance verification. Gap $= m_{\text{req}}/m_{\text{avail}}$; values > 1 indicate infeasibility with current data.

Framework	K	$\pi_{\min}$	δ	ε	m_{req}	m_{active}	m_{avail}	Gap
EU AI Act Annex III	10	0.05	0.020	0.050	1.25×10^6	2.50×10^4	7.07×10^5	1.8×
NIST AI RMF	5	0.10	0.010	0.030	1.50×10^6	1.50×10^4	5.00×10^5	3.0×
FDA SaMD	8	0.05	0.010	0.020	3.20×10^6	3.20×10^4	7.07×10^5	4.5×

where the last inequality uses $1 - \varepsilon \leq 1$ and the KL approximation $\mathrm{KL}(\mathrm{Bern}(p)\|\mathrm{Bern}(p+\delta)) \approx \delta^2/(p(1-p))$ (valid for $\delta \leq \varepsilon/2$). By Le Cam's lemma [Tsybakov, 2009]:

$$R^* \geq \frac{\delta}{2}\left(1 - \mathrm{TV}(P_0^m, P_1^m)\right) \geq \frac{\delta}{2} e^{-\mathrm{KL}(P_0^m \| P_1^m)}$$

where the second inequality uses Bretagnolle–Huber. Setting $\mathrm{KL} = 1$ (i.e., $\delta = \sqrt{\varepsilon/m}$): $R^* \geq (1/2e)\sqrt{\varepsilon/m} \approx 0.184\sqrt{\varepsilon/m}$. □

H.1 Direct Bernoulli Lower Bound (Alternative to Brown–Low)

For readers who prefer a self-contained argument avoiding asymptotic equivalence, we sketch a direct Assouad-type lower bound in the Bernoulli model.

Partition $[0, 1]$ into B equal-width bins. Consider the class of calibration functions $\Delta_\omega(p) = h \cdot \omega_b$ for $p \in I_b$, where $\omega = (\omega_1, \ldots, \omega_B) \in \{-1, +1\}^B$ and $h > 0$ is chosen so that Δ_ω is L-Lipschitz (requiring $h \leq L/(2B)$). Each Δ_ω has $\mathrm{ECE} = h$ (since $\sum_b (1/B)|h \cdot \omega_b| = h$). For any two configurations ω, ω' differing in one coordinate b, the KL divergence of the m/B observations in bin b is:

$$\mathrm{KL} = \frac{m}{B} \cdot \frac{(2h)^2}{\varepsilon(1-\varepsilon)} \leq \frac{4mh^2}{B\varepsilon}$$

By Assouad's lemma [Tsybakov, 2009], the minimax risk satisfies:

$$R^* \geq \frac{Bh}{2}\left(1 - \sqrt{\frac{2mh^2}{B\varepsilon}}\right)$$

Setting $h = c\sqrt{B\varepsilon/m}$ for a small constant c and $B = \lfloor (Lm/\varepsilon)^{1/3} \rfloor$ gives:

$$R^* \geq c' \left(\frac{L\varepsilon}{m}\right)^{1/3}$$

This recovers the $(L\varepsilon/m)^{1/3}$ rate directly from the Bernoulli model, confirming the Brown–Low reduction. The constant c' is weaker than the one obtained via Lepski et al. [1999], but the argument is fully self-contained. □

I Full Proof of Theorem 3 (Matched Upper Bound)

Bias-variance decomposition. With B equal-width bins and m total samples:

1. Bias. Within each bin of width $1/B$, the calibration gap Δ varies by at most L/B (Lipschitz). By the Bridge Lemma, $|\mathrm{ECE}_B - \mathrm{CE}_{\text{smooth}}| \leq L/B$.

2. Variance. Each bin b has $n_b \approx m/B$ samples. (Under bounded density $c_\mu \leq d\mu/dp \leq C_\mu$; equal-mass binning achieves $n_b = m/B$ exactly when μ concentrates.) The empirical calibration gap has variance $\mathrm{Var}(\hat{\Delta}_b) \leq \sigma^2/n_b \approx \sigma^2 B/m$, where $\sigma^2 = \max_b \mathbb{E}[\eta(p)(1-\eta(p)) \mid p \in I_b]$. Estimation error: $\mathbb{E}[|\widehat{\mathrm{ECE}}_B - \mathrm{ECE}_B|] \leq \sqrt{\sigma^2 B/m}$.

3. Total error. $L/B + \sqrt{\sigma^2 B/m}$.

4. Optimization. Setting $L/B = \sqrt{\sigma^2 B/m}$: $B^* = (L^2 m/\sigma^2)^{1/3}$, total error $(L\sigma^2/m)^{1/3}$. In the rare-error regime $\sigma^2 \approx \varepsilon$, recovering the headline rate. □

J Verification Dynamics, Recalibration Trap, and Half-Life

J.1 Verification Dynamics

Suppose a model is deployed at rate r samples per unit time and $\varepsilon(t) = c_0 t^{-\beta}$. The verification floor at time t is $\delta_{\text{floor}}(t) = (Lc_0/r)^{1/3} t^{-(\beta+1)/3}$. Meaningful verification ($\delta_{\text{floor}}(t) < \varepsilon(t)$) reduces to $(L/(rc_0^2))^{1/3} < t^{(1-2\beta)/3}$. This is satisfied for large t when $\beta < 1/2$, violated when $\beta > 1/2$, and equivalent to $L/(rc_0^2) < 1$ when $\beta = 1/2$. Critical exponent: $\beta^* = 1/2$ (passive), $\beta^*_{\text{active}} = 1$ (active). □

J.2 Recalibration Trap

With $\text{ECE}_K = \gamma^K \text{ECE}_0$, the improvement from round K to $K+1$ is $\gamma^K \text{ECE}_0(1-\gamma)$. Distinguishing requires (Theorem 1) $m_K \geq \varepsilon/(\gamma^{2K}\text{ECE}_0^2(1-\gamma)^2)$. Setting $m_K = M_{\text{total}}$ gives $K^* = \lfloor \log(M_{\text{total}}(1-\gamma)^2 \text{ECE}_0^2/\varepsilon)/(2\log(1/\gamma)) \rfloor$.

Worked example (MMLU). $\gamma = 0.5$, $\varepsilon = 0.05$, $\text{ECE}_0 = 0.10$, $M_{\text{total}} = 14{,}000$: $K^* = \lfloor 6.55/1.386 \rfloor = 4$. After 4 halving rounds, MMLU cannot verify further improvement.

J.3 Verification Half-Life

Let ECE drift rate λ satisfy $|\text{ECE}(t) - \text{ECE}(0)| \leq \lambda t$. (i) The verification becomes invalid after $t_{1/2} = \delta/\lambda$. (ii) Perpetual verification requires data rate $r \geq L\varepsilon\lambda/\delta^4$.

Domain estimates (assuming $L = 1$, $\varepsilon = 0.05$): Medical AI ($\lambda = 0.01$/month, $\delta = 0.02$): $t_{1/2} = 2$ months, $r \approx 3{,}100$/month. Content moderation: 6 months, 620/month. Financial trading ($\lambda = 0.1$/week, $\delta = 0.01$): $t_{1/2} = 17$ hours. Autonomous driving: 15 days. Static benchmarks have the same half-life: a benchmark constructed at $t = 0$ decays at rate λ, formalizing the case for periodic refresh.

K Alternative Derivation of Theorem 2 via Asymptotic Equivalence

Step 1: Reduction to nonparametric regression.

Within a window of width w around $p = 1 - \varepsilon$, where the score distribution concentrates, the calibration verification problem reduces to estimating the L_1 norm of a regression function. Specifically:

For $p \in [1 - \varepsilon - w/2,\ 1 - \varepsilon + w/2]$, define $\Delta(p) = \eta(p) - p$. The labels Y_i given $p(X_i) = p$ are $\text{Bern}(\eta(p)) = \text{Bern}(p + \Delta(p))$, with variance $\eta(p)(1-\eta(p)) \approx \varepsilon$ in the rare-error regime.

The ECE restricted to this window is:

$$\text{ECE}_{\text{window}} = \int_{1-\varepsilon-w/2}^{1-\varepsilon+w/2} |\Delta(p)|\, d\mu(p) \tag{2}$$

This is precisely the L_1 norm of Δ on the window, weighted by μ.

Step 2: Identification with L_1 functional estimation in white noise.

By the asymptotic equivalence of nonparametric regression and Gaussian white noise [Brown and Low, 1996, Nussbaum, 1996], estimating $\int |\Delta(p)| dp$ from m observations with per-observation variance ε is asymptotically equivalent to estimating $\int |f(t)| dt$ in the white noise model:

$$dX(t) = f(t)dt + (m/\varepsilon)^{-1/2} dW(t), \quad t \in [0,1] \tag{3}$$

with effective sample size $n_{\text{eff}} = m/\varepsilon$ and $f \in \Sigma(\beta = 1, L)$ (Lipschitz = Hölder with $\beta = 1$).

Step 3: Application of Lepski, Nemirovski & Spokoiny (1999).

By Theorem 2.2 of Lepski et al. [1999], the minimax rate for estimating the L_1 norm of a β-Hölder function in white noise with sample size n is:

$$R^*_{L_1} \geq c \cdot n^{-\beta/(2\beta+1)}/(\log n)^{c'} \tag{4}$$

For $\beta = 1$ (Lipschitz) and $n = n_{\text{eff}} = m/\varepsilon$:

$$R^*_{L_1} \geq c \cdot (m/\varepsilon)^{-1/3}/(\log(m/\varepsilon))^{c'} = c \cdot \left(\frac{\varepsilon}{m}\right)^{1/3} /(\log m)^{c'} \tag{5}$$

Incorporating the Lipschitz constant L (which scales the function class):

$$R^*(m, \varepsilon, L) \geq c_2 \left(\frac{L\varepsilon}{m}\right)^{1/3} /(\log m)^{c_3} \tag{6}$$

Technical conditions:

- *Score-mass concentration:* The score distribution μ must place mass $\geq \rho$ in a window of width w around $1 - \varepsilon$. We state this as an assumption. This holds for well-trained classifiers: overconfident models concentrate predictions near the top.
- *The Brown–Low equivalence* requires m sufficiently large relative to the smoothness. The minimum m is $m \geq C(L)/\varepsilon$ for a constant depending on L.
- *Lipschitz constant:* Since $\Delta(p) = \eta(p) - p$ and $p \mapsto p$ is 1-Lipschitz, we have $L_\Delta \leq L_\eta + 1$. We use L throughout for L_Δ.

Clarification on the asymptotic regime. The Brown–Low equivalence is applied in the limit $m \to \infty$ with ε *fixed*. The error rate ε is a property of the model under audit, not an asymptotic parameter. For any fixed $\varepsilon \in (0, 1/2)$, the Bernoulli label variance $\varepsilon(1 - \varepsilon) \geq \varepsilon/2 > 0$ is bounded away from zero, satisfying the regularity conditions. The ε-dependence in the rate $(L\varepsilon/m)^{1/3}$ arises from substituting $\sigma^2 = \varepsilon(1 - \varepsilon) \approx \varepsilon$ into the white-noise minimax rate with $n_{\text{eff}} = m/\varepsilon$, not from taking $\varepsilon \to 0$ within the equivalence framework.

L Supplementary Details for Theorem 11

The proof of Theorem 11 in the main text is complete. We note that the result extends straightforwardly to any side information $\mathcal{S}$ that is independent of the fresh evaluation labels conditional on the scores. This includes: model weights, architecture specifications, training data, training logs, gradient histories, and any derived quantity (e.g., Fisher information matrices, Hessian spectra, pruning masks). The key insight is that calibration is a property of the model-world interface, not of the model alone.

M Full Proof of Sequential Verification

Lower bound. Under the two Le Cam hypotheses (P_0: ECE $= 0$, P_1: ECE $= \delta$), the log-likelihood ratio after t observations is:

$$\Lambda_t = \sum_{i=1}^{t} \log \frac{p_1(y_i \mid p_i)}{p_0(y_i \mid p_i)} \tag{7}$$

This is a random walk with per-step drift:

$$\mathbb{E}_{P_0}[\Lambda_1] = -\mathrm{KL}(P_0(\cdot \mid p_i) \| P_1(\cdot \mid p_i)) \approx -\frac{\delta^2}{\varepsilon} \tag{8}$$

By Wald's identity for sequential testing [Tsybakov, 2009], any test between P_0 and P_1 with error probabilities $\leq \alpha$ satisfies:

$$\mathbb{E}[\tau] \geq \frac{\log(1/\alpha)}{\mathrm{KL}(P_0 \| P_1)} = \frac{\varepsilon \log(1/\alpha)}{\delta^2} \tag{9}$$

Since any sequential verification protocol with accuracy δ at confidence $1 - \alpha$ must distinguish ECE $= 0$ from ECE $= \delta$ with probability $\geq 1 - \alpha$, the lower bound follows.

Upper bound. We construct a confidence sequence for the ECE using the method of mixtures [Howard et al., 2021]. Define the running ECE estimate:

$$\widehat{\mathrm{ECE}}_t = \sum_{b=1}^{B} \frac{n_{b,t}}{t} \left|\hat{\Delta}_{b,t}\right| \tag{10}$$

where $\hat{\Delta}_{b,t}$ is the empirical calibration gap in bin b after t observations.

By Theorem 1 of Howard et al. [2021], there exists a time-uniform confidence sequence $(C_t)_{t\geq 1}$ such that:

$$\Pr\left(\exists t \geq 1 : |\widehat{\mathrm{ECE}}_t - \mathrm{ECE}| > C_t\right) \leq \alpha \tag{11}$$

with $C_t = O(\sqrt{\varepsilon \log(\log(t)/\alpha)/t})$.

Define the stopping time $\tau = \inf\{t : C_t \leq \delta\}$. Solving $C_t \leq \delta$:

$$\sqrt{\frac{\varepsilon \log(\log(t)/\alpha)}{t}} \leq \delta \implies t \geq \frac{\varepsilon \log(\log(t)/\alpha)}{\delta^2} \tag{12}$$

For the dominant term, $\mathbb{E}[\tau] = O(\varepsilon \log(1/\alpha)/\delta^2)$, matching the lower bound up to constants. □

N Temperature Scaling Analysis

We provide the full proof of the temperature scaling rate proposition.

Step 1: MLE for the temperature parameter. Given scores $p_1, \ldots, p_m$ and labels $y_1, \ldots, y_m$, the negative log-likelihood is:

$$\ell(T) = -\sum_{i=1}^{m} [y_i \log \sigma(\mathrm{logit}(p_i)/T) + (1 - y_i) \log(1 - \sigma(\mathrm{logit}(p_i)/T))] \tag{13}$$

This is a 1D convex optimization. The MLE $\hat{T}$ is unique.

Step 2: Asymptotic normality of MLE. By standard M-estimation theory [van der Vaart, 1998]:

$$\sqrt{m}(\hat{T} - T^*) \xrightarrow{d} N(0, \mathcal{I}(T^*)^{-1}) \tag{14}$$

where $\mathcal{I}(T^*)$ is the Fisher information. At $T^* = 1$, the score function is:

$$\frac{\partial}{\partial T} \log p(Y \mid p, T)\bigg|_{T=1} = -(Y - p) \cdot \mathrm{logit}(p) \tag{15}$$

So:

$$\mathcal{I}(1) = \mathbb{E}[(Y - p)^2 \cdot \mathrm{logit}(p)^2] = \mathbb{E}[p(1 - p) \cdot \mathrm{logit}(p)^2] \tag{16}$$

In the rare-error regime with scores concentrated near $1-\varepsilon$: $\mathrm{logit}(1-\varepsilon) \approx \log(1/\varepsilon)$ and $p(1-p) \approx \varepsilon$, giving:

$$\mathcal{I}(1) \approx \varepsilon \cdot \log^2(1/\varepsilon) \tag{17}$$

This is positive and grows as $\varepsilon \to 0$ — rare errors make temperature estimation *easier* because logits are more spread out. The MLE satisfies: $|\hat{T} - T^*| = O_p(1/\sqrt{m \cdot \mathcal{I}(T^*)})$.

Step 3: Translating to calibration error. By Taylor expansion of $\Delta(p; T)$ around $T^* = 1$:

$$\Delta(p; \hat{T}) \approx \frac{\partial \Delta}{\partial T}\bigg|_{T=1} \cdot (\hat{T} - 1) = p(1 - p) \cdot \mathrm{logit}(p) \cdot (\hat{T} - 1) \tag{18}$$

So the smooth CE is:

$$\mathrm{CE}_{\mathrm{smooth}} = \int |\Delta(p; \hat{T})| \, d\mu(p) \approx |\hat{T} - 1| \cdot \underbrace{\int p(1 - p)|\mathrm{logit}(p)| \, d\mu(p)}_{\kappa} \tag{19}$$

where κ is a constant depending on the score distribution. Substituting $|\hat{T}-1| = O_p(1/\sqrt{m \cdot \mathcal{I}(T^*)})$:

$$\mathrm{CE}_{\mathrm{smooth}} = O_p\left(\frac{\kappa}{\sqrt{m \cdot \mathcal{I}(T^*)}}\right) = O_p\left(\frac{1}{\sqrt{m}}\right) \tag{20}$$

with a constant that depends on ε through $\mathcal{I}(T^*)$ and κ. This is the **parametric rate** $m^{-1/2}$, not the nonparametric rate $(L\varepsilon/m)^{1/3}$.

Step 4: Connection to binned ECE. Via the Bridge Lemma: $\mathrm{ECE}_B \leq \mathrm{CE}_{\mathrm{smooth}} + L/B$. Taking $B = O(\sqrt{m}/L)$:

$$\mathrm{ECE}_B = O(1/\sqrt{m}) \tag{21}$$

Comparison to nonparametric rate: The ratio is $(1/\sqrt{m})/(L\varepsilon/m)^{1/3} = (L\varepsilon)^{-1/3}/m^{1/6}$, which shrinks with m. For $m = 10{,}000$ and $L\varepsilon = 0.05$: ratio ≈ 0.05. Temperature scaling is $\sim$20$\times$ more data-efficient than nonparametric ECE estimation. □

O Verification Transfer Details

We provide the full proof of the verification transfer proposition.

Step 1: Problem reduction. Since $\Delta_1(p)$ is known (from prior verification), estimating $\mathrm{ECE}(M_2) = \int |\Delta_2(p)|\, d\mu(p)$ reduces to estimating the shift $h(p) = \Delta_2(p) - \Delta_1(p)$, then computing $\int |\Delta_1(p) + \hat{h}(p)|\, d\mu(p)$.

Step 2: Lower bound (detection cost). Two-sample Le Cam between H_0: $\Delta_2 = \Delta_1$ and H_1: $\Delta_2 = \Delta_1 + h$ with $\|h\|_\infty = \delta_{\mathrm{ft}}$. From m_2 fresh samples near $p \approx 1 - \varepsilon_2$:

$$\mathrm{KL}(H_0^{m_2} \| H_1^{m_2}) \approx \frac{m_2 \delta_{\mathrm{ft}}^2}{\varepsilon_2} \tag{22}$$

Indistinguishable when $\mathrm{KL} \leq 1$, requiring $m_2 = \Omega(\varepsilon_2/\delta_{\mathrm{ft}}^2)$.

Step 3: Upper bound (estimation with transfer). Apply Theorem 3 to h instead of Δ_2. The histogram estimator for h has bias L_h/B and variance $\sqrt{\varepsilon_2 B/m_2}$. Optimizing: $B^* = (L_h^2 m_2/\varepsilon_2)^{1/3}$, giving:

$$\|\hat{h} - h\| = O\left(\left(\frac{L_h \varepsilon_2}{m_2}\right)^{1/3}\right) \tag{23}$$

By the triangle inequality:

$$\left|\mathrm{ECE}(M_2) - \widehat{\mathrm{ECE}}(M_2)\right| \leq \int |\hat{h}(p) - h(p)|\, d\mu(p) = O\left(\left(\frac{L_h \varepsilon_2}{m_2}\right)^{1/3}\right) \tag{24}$$

Setting this equal to δ_{ft}: $m_2 = O(L_h \varepsilon_2/\delta_{\mathrm{ft}}^3)$.

Step 4: When does transfer help? From-scratch: $m_2 = \Theta(L\varepsilon_2/\delta^3)$. With transfer: $m_2 = \Theta(L_h \varepsilon_2/\delta_{\mathrm{ft}}^3)$. Transfer helps when $\delta_{\mathrm{ft}} < \delta$ and $L_h \leq L$. With $\delta_{\mathrm{ft}} = \delta/10$, saving is $1000\times$. Transfer provides no benefit when $\delta_{\mathrm{ft}} \geq \delta$. □

P Corrections: Correlation, Heterogeneity, Pre-Calibration

The core theorems assume i.i.d. labeled samples with independent Bernoulli noise. In practice, several deviations may arise.

Temporal correlation. If evaluation samples arrive sequentially and exhibit temporal dependence, the effective sample size is reduced by a factor depending on the mixing time. For β-mixing sequences with coefficient $\beta(k) \leq c \cdot \rho^k$, the effective sample size is $m_{\mathrm{eff}} \approx m \cdot (1 - \rho)$. All bounds apply with m replaced by m_{eff}.

Domain heterogeneity. When the evaluation set spans multiple domains with different error rates, the per-domain verification floor (Fix 3) accounts for this heterogeneity. The overall ECE is a weighted average of domain-specific ECEs, and Jensen's inequality ensures the overall verification floor is no larger than the worst per-domain floor weighted by domain proportion.

Pre-calibration via RLHF. Models that have undergone reinforcement learning from human feedback may have artificially smooth calibration functions (small L), which reduces the verification tax. However, if the RLHF procedure oversmooths, the remaining miscalibration may be concentrated at specific confidence levels, violating the uniform Lipschitz assumption. Domain-specific Lipschitz estimates should be used in such cases.

Q Numerical Sanity Checks

Q.1 Sharp Le Cam Constants

For each ε, we numerically optimize the Le Cam lower bound over the separation parameter δ, using the exact Bernoulli KL divergence (not the quadratic approximation). We compute three versions: the Bretagnolle–Huber bound $R^* \geq (\delta/2)\exp(-\mathrm{KL})$, the Pinsker bound $R^* \geq (\delta/2)(1-\sqrt{\mathrm{KL}/2})$, and the exact bound via the Bernoulli likelihood ratio test. Table 8 reports the tightest constant for each ε.

Table 8: Sharp Le Cam constants $c_1(\varepsilon)$ and nonparametric estimation sample sizes. The Le Cam constant governs detection; the estimation column uses $m \geq L\varepsilon/\delta^3$ with $L = 1$, $\delta = 0.01$.

ε	$c_1(\varepsilon)$	m_{detect} (Le Cam)	m_{estimate} (nonparametric)
0.01	0.3780	15	10,000
0.02	0.3638	27	20,000
0.05	0.3474	61	50,000
0.10	0.3326	111	100,000
0.15	0.3206	155	150,000
0.20	0.3093	192	200,000
0.25	0.2982	223	250,000
0.30	0.2873	248	300,000
0.35	0.2759	267	350,000

The constant $c_1(\varepsilon)$ is monotonically decreasing in ε: smaller error rates yield tighter constants because the Bernoulli variance is smaller and the KL approximation is sharper. All values fall in $[0.28, 0.38]$, consistent with the theoretical prediction $c_1 \geq 1/(2e) \approx 0.184$ from the quadratic KL approximation. The exact constants are roughly $2\times$ larger, reflecting the tightness gained from exact Bernoulli KL computation.

R Synthetic Experiments

Setup. Calibration gap $\Delta(p) = A\sin(2\pi kp)$ with $L = 2\pi kA$. Scores from $\mathrm{Beta}((1-\varepsilon)/\varepsilon, 1)$. Labels $Y \sim \mathrm{Bern}(p + \Delta(p))$. Optimal B^*.

Results. (1) Phase transition validates at $m\varepsilon \approx 1$ (Fig. 2). (2) Passive slope converges to $-1/3$ as k increases (zero-crossings drive worst-case). (3) Active-passive gap widens with L.

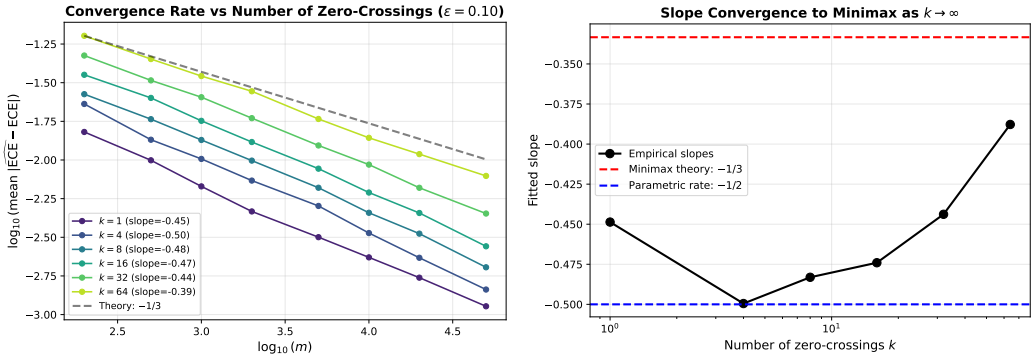

Figure 9: Slope convergence to minimax $-1/3$ as zero-crossings increase (synthetic).

S Real-Model Experiment Details

MMLU experimental protocol. We query each NIM-hosted model with the standard 4-way multiple-choice prompt ("Answer the following multiple choice question. Reply with ONLY the

letter (A, B, C, or D)"). For each question, we extract the logprobs of the four answer tokens A, B, C, D at the first generated position, apply softmax over those four logits, and take the maximum as the model's confidence. The model's predicted answer is the argmax. Records with API errors (e.g., 403 Forbidden, content-filter rejection) are excluded. Each model run uses a single deterministic temperature-0 generation per question.

Confidence saturation (full data, $N = 14{,}042$ each). On the complete MMLU set, max-softmax confidence is heavily concentrated near 1: Llama-3.1-405B has mean confidence 0.96 with 77% of questions saturated (> 0.99); Qwen3-Next-80B has mean 0.98 with 86% saturated; Llama-4-Maverick-17B has mean 0.99 with 92% saturated. The 405B model has the most spread (lowest saturation rate), as expected for a larger and more capable model. Saturation affects the resolution of high-confidence bins but not the convergence rate, which is governed by the optimal bin count B^*.

Lipschitz constant estimation. For each model, we partition the score range into 20 equal-width bins, compute the empirical accuracy in each bin with at least 30 samples, take the calibration gap $\hat{\Delta}_b = \mathrm{acc}_b - p_b$ at each bin center, and form the adjacent-bin slope $|\hat{\Delta}_{b+1} - \hat{\Delta}_b|/|p_{b+1} - p_b|$. We report the 75th percentile of these slopes, capped at $\hat{L} = 5$, as a robust estimate. This is sensitive to bin choice and noise; we recommend the audit floor be reported with $\hat{L}$ from at least two binning resolutions.

Pseudo-classifier control experiment. As a sanity check before running the real-model experiment, we constructed a 10-class softmax classifier from Gaussian logits with $\varepsilon \approx 0.10$ ($N = 20{,}000$, full-dataset ECE ≈ 0.31). Subsampling at $m \in \{50, \ldots, 10^4\}$ with optimal B^* bins (200 replicates per m) confirms that (i) the verification floor (red dashed in Figure 10) tracks the empirical std of estimates across the full range, and (ii) the phase transition near $m \cdot \varepsilon \approx 1$ is visible. The pseudo-classifier results match the MMLU real-model findings of Section 8.

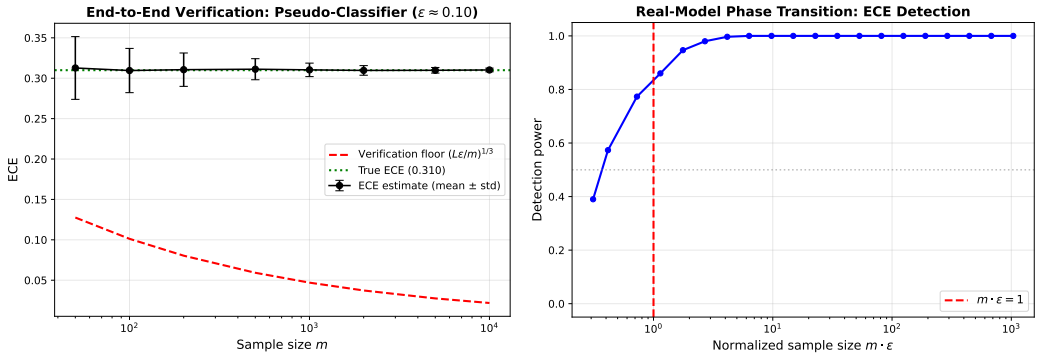

Figure 10: Pseudo-classifier control experiment ($\varepsilon \approx 0.10$, true ECE ≈ 0.31, $N = 20{,}000$). (Left) ECE estimate $\pm$ std vs. m with verification floor overlaid. (Right) Detection power vs. $m \cdot \varepsilon$, with the predicted phase transition at $m \cdot \varepsilon = 1$.

S.1 Named Model Comparisons

Full methodology: pairwise accuracy gaps between 10 frontier models on 4 benchmarks using published scores from technical reports and model cards. Verification floor: $\delta_{\mathrm{acc}} = 2\sqrt{\varepsilon(1-\varepsilon)/n}$. The main-text Table 4 shows the top comparisons; full data (64 pairs) is in the supplementary materials.

T Full Proof of Theorem 13 (Compositional Verification)

Lipschitz composition. By the chain rule, if g_k is L_k-Lipschitz in its first argument, then $g_K \circ \cdots \circ g_1$ is $(\prod_{k=1}^{K} L_k)$-Lipschitz. Since $\Delta_{\mathrm{sys}}(p) = \eta_{\mathrm{sys}}(p) - p$ and $p \mapsto p$ is 1-Lipschitz, $L_{\mathrm{sys}} \leq \prod_k L_k + 1$.

Verification cost. Applying Theorem 3 with $L = L_{\text{sys}}$: $m_{\text{sys}} = \Theta(L_{\text{sys}}\varepsilon/\delta^3)$. For homogeneous $L_k = L$: $L_{\text{sys}} \approx L^K$, giving exponential scaling. □

Agent loop bound. For an agent with K reasoning iterations and $L = 2$: $L^{10} = 1024$, $L^{20} > 10^6$.

Caveats. The Lipschitz composition model is optimistic in two directions: discontinuous components (e.g., discrete retrieval) yield $L_k = \infty$, making end-to-end verification impossible at *any* sample size; conversely, contractive components ($L_k < 1$) reduce the effective L_{sys}. Empirical estimation of L_{sys} via finite-difference approximation is recommended before applying the exponential bound. Independence of per-component calibration gaps is also assumed; correlated miscalibration can either help or hurt depending on structure.

Worked example. A two-stage retriever–classifier with $L = 2$ each gives $L_{\text{sys}} \leq 5$, requiring $5\times$ the data of a single classifier. With $L = 3$: $10\times$. A three-stage pipeline at $L = 2$: $9\times$; at $L = 3$: $28\times$.